

PASSIVE RECOVERY IS SUPERIOR TO ACTIVE RECOVERY DURING A HIGH-INTENSITY SHOCK MICROCYCLE

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¹*Institute of Training Science and Sport Informatics, German Sport University, Cologne, Germany;* ²*Department of Molecular and Cellular Sport Medicine, Institute of Cardiovascular Research and Sport Medicine, German Sport University, Cologne, Germany;* and ³*The German Research Center of Elite Sport, German Sport University, Cologne, Germany*

ABSTRACT

Wahl, P, Zinner, C, Grosskopf, C, Rossmann, R, Bloch, W, and Mester, J. Passive recovery is superior to active recovery during a high-intensity shock microcycle. *J Strength Cond Res* 27(5): 1384–1393, 2013—The purpose was to examine the effects of a 2-week high-intensity shock microcycle on maximal oxygen consumption and parameters of exercise performance in junior triathletes on the one hand and to evaluate the long-term effects of active (A) vs. passive (P) recovery on the other hand. Sixteen healthy junior triathletes participated in the study. For the assignment to the A or P group, the subjects were matched according to age and performance. Within 2 weeks, a total of 15 high-intensity interval sessions within three 3-day training blocks were performed. Before and 1 week after the last training session, the athletes performed a ramp test to determine $\dot{V}O_{2\max}$, a time trial (TT) and a Wingate test. Furthermore, total hemoglobin (Hb) mass was determined. The results of the whole group, independent of the arrangement of recovery, were analyzed at first; second, the A and P groups were analyzed separately. Peak power output (PPO) during the ramp test and TT performance significantly increased in the whole group. The comparison of the 2 groups revealed increases for the mentioned parameters and for $\dot{V}O_2$ and power output at VT2 for the P group only. The $\dot{V}O_{2\max}$ did not change. Wingate performance increased in the A group only. The tHb mass slightly decreased. The main finding of this study was that a 14-day shock microcycle is able to improve TT performance and PPO in junior triathletes in a short period of time. Furthermore, not only the intensity but also the arrangement of interval training seems to be important as well, because only the P group showed improvements in endurance performance, despite a slightly lower training volume.

These findings might be relevant for future arrangements of high-intensity interval training.

KEY WORDS block periodization, triathlon, time trial, $\dot{V}O_{2\max}$, Wingate, tHb mass

INTRODUCTION

Triathlon is a unique endurance sport that comprises a sequential swim, cycle, and run over a variety of 'short' or 'long' distances. Except of variations between the different distances the main energy demand is covered by aerobic energy supply. However, the analysis of the intensity zones during cycling of an Olympic distance triathlon (40 km) show, that approximately 19% of the total cycling time are $>100\%$ of the maximal aerobic power and approximately 14% are between ventilator threshold 2 (VT2) and 100% of the maximal aerobic power respectively, showing a large percentage of high intensities (13). The primary determinant of success is the ability to sustain a high rate of (aerobic) energy expenditure/capacity for prolonged periods of time. Therefore a high $\dot{V}O_{2\max}$ in individuals is clearly of high importance to triathlon performance showing ranges of 52–70 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ in men and 59–66 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ in women (26). Consequently, the improvement of $\dot{V}O_{2\max}$ is one important aim to improve performance. However, other factors such as the economy of motion and the fractional use of $\dot{V}O_{2\max}$ determine performance as well (28).

Previous studies have shown that endurance-related variables, such as $\dot{V}O_{2\max}$ and anaerobic threshold, are difficult to enhance in well-trained athletes through additional increases in submaximal training volume (6). In contrast, several studies indicate that high-intensity interval training (HIT) improves endurance capacity and $\dot{V}O_{2\max}$ more effectively than submaximal training in sedentary and recreationally active individuals and in elite athletes (37,38). The $\dot{V}O_{2\max}$ was improved by 4.9–8.1% in well-trained cyclists and runners after HIT performed 2–3 times per week for a 3- to 8-week period (22,33). Thus, it seems that for trained athletes, training at or very close to $\dot{V}O_{2\max}$ is an effective

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stimulus to further enhance $\dot{V}O_{2\max}$ (21,25). In most of the studies (with elite athletes), 1–4 HIT sessions per week were performed (37). Seiler and Tonnessen (32) and Billat et al. (1) suggested 2–3 HIT session per week as an agreeable amount and optimum for elite endurance athletes, respectively (1,32). Despite these recommendations, still little is known about the integration of HIT in daily training regimes and its usage during various training periods.

Another concept for the application of HIT, in comparison with the periodical integration of single HIT sessions, are high-intensity shock microcycles with up to 10 HIT sessions per week during a 2-week period (2,17). By using these short-term shock microcycles, it is possible to apply a high stimulus and to improve endurance performance as previous studies were able to show (2,34). However, only these 2 studies are available and were performed in technical sport disciplines such as football and alpine skiing, but up to now, no study investigated the effects of a shock microcycle in endurance sports, such as triathlon.

Besides the question of the integration of HIT into the training cycle, little is known about the effects of different arrangements of interval training. Questions remain concerning the intensity, duration and amount of intervals, and the arrangement of recovery between intervals (e.g., active or passive recovery) (21,22). Only a few studies investigated the (acute) effects of different intensities during HIT on aerobic capacity and performance (8,15). No study looked for the long-term effects of different recovery arrangements (active vs. passive recovery). Previous studies were able to show that lactate and pH are potent metabolic stimuli inducing adaptations and hormonal responses (29) and whose changes, because of physical activity, vary depending on the intensity of the interval and the arrangement of recovery (faster normalization of pH and lactate levels because of active recovery) (12). Passive recovery, compared with active recovery, between HIT intervals might lead to greater and more long-lasting metabolic alterations, which were shown to be essential for muscle adaptation in endurance training (4) and improvements of endurance performance. Therefore, the

arrangement of recovery might be as important as the intensity and duration of the interval itself.

According to the aforementioned data the aim of this study was to test the effect of high-intensity training according to block periodization and the effects of active vs. passive recovery on the aerobic capacity in junior triathletes. We hypothesized (a) that a 14-day shock microcycle including 15 HIT sessions elicits a great improvement in $\dot{V}O_{2\max}$ and time trial (TT) performance in triathletes and (b) that the arrangement of interval training (active vs. passive recovery) would elicit different results. We therefore analyzed the results of the whole group, independent of the arrangement of recovery and the active (A) and passive (P) recovery group separately.

METHODS

Experimental Approach to the Problem

To test the first hypothesis, the athletes performed a 14-day HIT shock microcycle. With the aim of testing the second hypothesis, the subjects were divided into 2 groups; one performing active recovery between the intervals of an HIT session, the other performing passive rest. For the assignment to the A or P group, the subjects were matched according to age and performance. The investigation was conducted during the athletes' (off-season) preparatory period. Before and 7 days after the intervention (to assure sufficient recovery [2]), performance tests were carried out. All the tests were performed under controlled conditions in our laboratory. The athletes were instructed to avoid strenuous exercise for 24 hours before all testing.

Subjects

Sixteen healthy competitive junior triathletes from a local team (12 male, 4 female) participated in this study. Before the investigation started, the athletes used to train for approximately 3 months, in which the amount of training was increased successively. The weekly training amount was approximately 15–25 h · wk⁻¹, mainly consisting of HVT. All the athletes had at least 2 years of training experiences. Anthropometric data are shown in Table 1. All the subjects

TABLE 1. Characteristics and anthropometric data of the subjects before (pre) and 7 days after (post) the training intervention.*

	Overall		Active		Passive	
	Pre	Post	Pre	Post	Pre	Post
Age (y)	15.4 ± 1.9	15.4 ± 1.9	15.1 ± 2.0	15.1 ± 2.0	15.6 ± 1.9	15.6 ± 1.9
Height (cm)	172.5 ± 9.5	172.5 ± 9.5	170.4 ± 9.1	170.4 ± 9.1	174.6 ± 10.0	174.6 ± 10.0
Body mass (kg)	57.7 ± 8.9	58.3 ± 8.9†	56.7 ± 10.6	57.6 ± 10.4†	58.7 ± 7.5	59.0 ± 7.8

*Parameters are shown as mean ± SD. Overall: analysis of the whole group independent of the arrangement of recovery ($n = 16$); active and passive group: $n = 8$ each.

†Significantly different from prevalences.

and guardians gave written informed consent to the contribution to the study and the study protocol was performed in accordance with the declaration of Helsinki and the Ethical Committee of the university.

Procedures

During the shock microcycle, a total of 15 high-intensity interval sessions within three 3-day training blocks were performed. Training blocks were separated by 1 rest day each (Figure 1) where only technical drills in swimming were performed. The majority of the HIT sessions (10) were performed on a cycle ergometer (Cyclus 2; RBM elektronik-automation GmbH, Leipzig, Germany), consisted of four 4-minute interval bouts at an intensity eliciting 90–95% of the individual maximal heart rate (HR_{max}), separated by 3-minute active recovery (60% of the individual HR_{max}) or passive recovery periods. However, to meet the demands of all 3 disciplines and to break the monotony, HIT sessions were also performed in running (3) and swimming (2) (Figure 1). In running HIT sessions consisted of 2×10 40-second interval bouts at an intensity eliciting 90–95% of the individual HR_{max}, separated by 20-second active (60% of the individual HR_{max}) or passive recovery periods. After the first 10 repetitions, the subjects had a 3-minute break (active or passive). In swimming, the subjects performed six 200-m swims (~90% of seasonal personal best time) separated by 2-minute active (~50% of seasonal personal best time) or passive recovery periods. Training intensity for cycling and running was controlled by continuous HR monitoring. The HIT sessions were supervised and performed in groups of 6 athletes (cycling). A standardized continuous 10-minute warm-up was performed before all HIT sessions in each discipline.

After the arrival in the morning, anthropometric data (weight, height, body fat) were gathered (Tanita BC-418 MA, Tanita Europe B.V., Amsterdam, The Netherlands). The subjects performed 3 performance tests within 2 days in the same order and at same time of day. Food intake and hydration was partially standardized. That is, certain foods containing carbohydrates were recommended to the sub-

jects and the subjects were advised to ingest the same amount and composition of food and drinks before each test. Therefore, food-drink intake was recorded before the first test and reproduced before the following. The subjects were not allowed to perform strenuous exercise 24 hours before testing. The first day, the triathletes performed a ramp test and 5 hours later a TT. The next day, a Wingate Anaerobic Test was conducted, followed (8 hours later) by the measurements of blood volume (BV) and total hemoglobin mass (tHb mass). Before the pretesting, all the subjects were fully familiarized with the performance tests.

The ramp test was performed on a cycle ergometer (Schoberer Rad Meßtechnik SRM GmbH, Jülich, Germany) to determine $\dot{V}O_{2\max}$ and peak power output (PPO). Breath-by-breath respiratory data were collected using a calibrated spirometry system (Zan 600, Zan Messgeräte, Oberthulba, Germany). Before each test, the volume transducer was calibrated using a 1-L calibration syringe and the CO_2 and O_2 gas analyzers were calibrated automatically using a known gas mixture. The ramp test consisted of a 10-minute warm-up at $1 \text{ W} \cdot \text{kg}^{-1}$ body weight for female subjects and $1.5 \text{ W} \cdot \text{kg}^{-1}$ body weight for male participants followed by a 5-minute rest period. Afterward, the ramp started with 2 minutes at $1.5 \text{ W} \cdot \text{kg}^{-1} / 2.0 \text{ W} \cdot \text{kg}^{-1}$ body weight followed by an increase of 20 W every 30 seconds until exhaustion. All respiratory data were averaged every 30 seconds. The highest values for oxygen uptake within the last 30 seconds of the test were used for statistical analysis. The mean duration of the ramp test was 8.9 ± 0.8 minutes (24). During the whole test, heart rate was measured continuously (Polar S810, Polar Electro GmbH, Büttelborn, Germany).

To test sport-specific changes in performance, the subjects performed a 20-minute TT, as TT appear to be the most valid and reliable simulation of a race event (coefficient of variation <5%) (7). The TT consisted of a 10-minute warm-up at $1 \text{ W} \cdot \text{kg}^{-1}$ body weight for female participants and $1.5 \text{ W} \cdot \text{kg}^{-1}$ body weight for male participants followed by a 5-minute rest period and a 20-minute TT. Therefore, the cycle ergometer (Schoberer Rad Meßtechnik SRM GmbH, Jülich, Germany) was adjusted to an isokinetic mode,

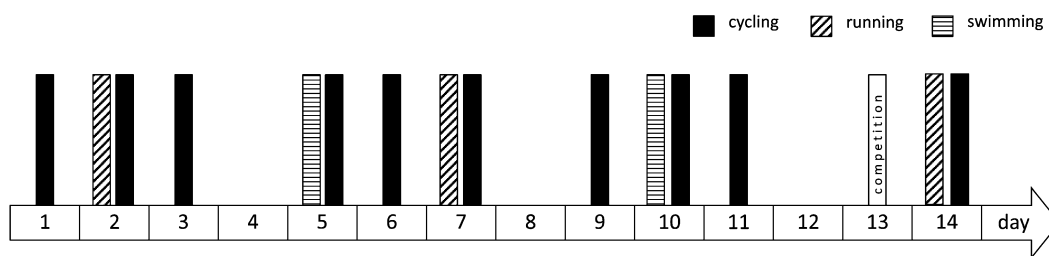


Figure 1. Arrangement of the 15 high-intensity interval training (HIT) sessions over the 11-day shock microcycle.

whereby the subjects were allowed to choose their favorite pedaling frequency. The same frequency was adjusted during the postintervention TT. Afterward, mean power output (MPO) was calculated. According to Currell and Jeukendrup (7), the subjects had no feedback of performance either during or after the trial until all the trials were completed, no distractions or encouragement during performance trials were given and physiological measurements were reduced to continuous HR monitoring (Acentas GmbH, Hörgertshausen, Germany) and lactate measurements (7). For lactate determination, 20 μ l of blood were withdrawn from the earlobe (pre, 5, 10, 15, and 20 minutes during the TT) and directly mixed with 1 ml of the EBIO plus system solution and measured with EBIOplus (EKF Diagnostic Sales, Magdeburg, Germany).

On the second day, a 30 seconds all-out exercise test (Wingate) was performed in an isokinetic mode set to a cadence of 120 rpm (warming-up see TT). The subjects were instructed to perform the tests in a sitting position on the ergometer. During the test, all the subjects were vocally encouraged to achieve maximal power output. Afterward, the peak power (PP), MPO, and the fatigue index (FI) (power decline) were calculated. The FI was calculated by the following formula: $FI = [(Peak\ Power - Lowest\ Power) \times 100] \times Peak\ Power^{-1}$. Capillary samples from the earlobe were collected pre and in minute intervals (1'–10') posttest to determine the “maximal rate of lactate accumulation (dLa/dt_{max})” according to Heck et al. (14) using the following formula: dLa/dt_{max} (millimoles per liter per second) = $([La]_{max} - [La]_{rest}) \times (t_{exerc.} - t_{alac})^{-1}$, where $[La]_{max}$ (millimoles per liter) = maximal lactate concentration after the exercise; $[La]_{rest}$ (millimoles per liter) = lactate concentration before exercise; $t_{exerc.}$ (seconds) = duration of exercise; t_{alac} (seconds) = period at the beginning of exercise for which (fictitiously) no lactate formation is assumed. The T_{alac} was set individually as the time to PP (seconds).

The BV and total hemoglobin mass (tHb mass) were measured according to the “optimized CO rebreathing method” as described by Gore et al. (11). The administered amount of CO in milliliters was individually calculated (ml CO = $0.8 \times$ body weight [kilograms, male participants] and milliliters CO = $0.7 \times$ body weight [kilograms, female participants], respectively). The HbCO amount in blood samples was measured with Radiometer OSM 3 (Copenhagen, Denmark).

On 4 separate days during the 2-week training period (days 1, 5, 10, 14), a scale was used to assess a person's perceived physical state (PEPS). The subjects were asked to judge spontaneously to what extent 20 given adjectives coincide with their current physical feeling. The adjective list PEPS was developed especially for young athletes (18). The adjectives of the PEPS represent 4 dimensions, each including 5 adjectives: perceived physical energy, perceived physical fitness, perceived physical flexibility, and perceived

physical health. The second part of the questionnaire consists of a short form of the “Eigenzustandsskala” (EZ scale) (27). As opposed to other psychological adjective scales (e.g., the POMS) the EZ scale measures not only the emotional or psychological strain but also the motivational state. The adjectives of the EZ scale represent 8 dimensions: positive mood, calmness, recovery, and sleepiness. The remaining 4 dimensions characterize the motivational state of a person: willingness to seek contact, social acceptance, readiness to strain, and self-confidence.

Statistical Analyses

Statistical analyses of the data were performed by using a statistics software package (Statistica for Windows, 7.0, Statsoft, Tulsa, OK, USA). Descriptive statistics of the data are presented as mean $\pm$ SD. Furthermore, 95% confidence interval (CI) is given. To test the hypothesis, if a 14-day shock microcycle elicits an improvement in physiological measures and performance, prevalues and postvalues of the whole group and of the A and P groups were compared separately using a paired *t* test.

To test our second hypothesis that the arrangement of interval training (active vs. passive recovery) would elicit different results the absolute changes of both groups (Δ post-pre) were compared with an unpaired *t*-test. Statistical differences were considered to be significant for $p \leq 0.05$. The effect size Cohen's *d* (defined as [difference between the means]/SD) was calculated for the comparison of prevalues and postvalues of each group (5). The thresholds for small, moderate, and large effects were defined as 0.20, 0.50, and 0.80, respectively, and the effect sizes are documented in Table 3.

Reliability (Cronbach's- α) and intraclass correlations (ICCs; 2-way random single measures) were calculated using IBM SPSS Statistics 20 (IBM Deutschland GmbH, Ehningen, Germany). Internal consistency was set as follows: $\alpha \geq 0.9$ excellent; $0.9 > \alpha \geq 0.8$ good; $0.8 > \alpha \geq 0.7$ acceptable; $0.7 > \alpha \geq 0.6$ questionable; $0.6 > \alpha \geq 0.5$ poor; $0.5 < \alpha$ unacceptable. Measurement of the power output on 2 different days before the study revealed a technical error (%TEM) of 1.1%. The TEM for oxygen uptake was 2.7%. Under our laboratory conditions, the coefficient of variation for repeated measurements of blood lactate concentration is routinely 1.1% at 12 mmol $\cdot$ L⁻¹.

RESULTS

Body mass increased significantly from pre to post in the whole and in the A group (Table 1). No significant changes were observed for the P group. Mean cycling work load during HIT sessions was 192 ± 44 W (range 150–300 W) for the whole group, 193 ± 51 W (range 150–300 W) for the A group, and 191 ± 42 W (range 140–270 W) for the P group.

The $\dot{V}O_{2max}$ relative to body mass showed no significant changes, either in the whole group, or in the A/P group (Cronbach's $\alpha = 0.95$; ICC: 0.9). The $\dot{V}O_2$ and power output

TABLE 2. Results from the ramp test, the time trial and the Wingate test before (pre) and 7 days after (post) the training intervention.*†

	Overall			Active		
	Pre	Post	<i>d</i>	Pre	Post	<i>d</i>
Ramp test						
PPO (W)	301 ± 49 (275–327)	315 ± 48‡ (289–340)	0.29	299 ± 58 (250–347)	311 ± 56 (§) (265–358)	0.21
PPO (W·kg ⁻¹)	5.2 ± 0.4 (5.0–5.5)	5.4 ± 0.5‡ (5.2–5.7)	0.44	5.3 ± 0.4 (4.9–5.6)	5.4 ± 0.5 (5.0–5.85)	0.22
ṀO ₂ max (ml·kg ⁻¹ ·min ⁻¹)	57.3 ± 8.5 (52.3–61.9)	57.9 ± 8.6 (53.3–62.4)	0.02	58.2 ± 9.3 (50.4–65.9)	57.9 ± 9.1 (50.3–65.4)	0.19
VT 1 _{power} (W)	184 ± 45 (160–208)	190 ± 37 (171–210)	0.15	191 ± 53 (147–236)	190 ± 43 (154–226)	0.02
VT 2 _{power} (W)	242 ± 52 (214–270)	248 ± 44 (224–271)	0.12	253 ± 64 (199–306)	245 ± 51 (202–288)	0.14
VT 1_ṀO ₂ (ml·min ⁻¹)	2.19 ± 0.63 (1.86–2.53)	2.27 ± 0.58 (1.96–2.58)	0.13	2.14 ± 0.82 (1.46–2.83)	2.14 ± 0.61 (1.63–2.65)	0
VT 2_ṀO ₂ (ml·min ⁻¹)	2.66 ± 0.69 (2.29–3.03)	2.84 ± 0.54 (2.56–3.13)	0.29	2.65 ± 0.86 (1.93–3.37)	2.69 ± 0.59 (2.20–3.19)	0.05
HR _{max} (b·min ⁻¹)	197 ± 7 (194–201)	196 ± 7‡ (196–199)	0.14	197 ± 8 (190–203)	195 ± 7 (189–201)	0.27
Time trial						
TT (W)	170 ± 41 (148–191)	190 ± 49‡ (165–217)	0.44	170 ± 45 (133–208)	185 ± 55 (§) (139–231)	0.3
TT (W·kg ⁻¹)	2.9 ± 0.4 (2.7–3.2)	3.2 ± 0.6‡ (2.9–3.5)	0.59	3.0 ± 0.3 (2.71–3.26)	3.1 ± 0.5 (2.74–3.56)	0.24
HR _{TT} (b·min ⁻¹)	182 ± 8 (177–186)	180 ± 7 (176–184)	0.27	180 ± 10 (172–189)	179 ± 7 (172–185)	0.12
Lactate (mmol·L ⁻¹)	6.2 ± 0.8 (5.4–7.1)	7.4 ± 1.1‡ (6.0–8.7)	1.25	5.9 ± 0.9 (4.8–7.0)	6.6 ± 1.5 (4.7–8.5)	0.57
Wingate						
PPO (W)	731 ± 189 (630–832)	729 ± 181 (633–826)	0.01	685 ± 221 (501–870)	698 ± 222 (512–884)	0.06
PPO (W·kg ⁻¹)	12.5 ± 1.6 (11.7–13.4)	12.4 ± 1.6 (11.5–13.3)	0.06	11.9 ± 1.9 (10.3–13.5)	11.9 ± 1.7 (10.5–13.3)	0
Time to PPO (s)	5.3 ± 1.1 (4.8–5.9)	5.3 ± 1.0 (4.7–5.8)	0	5.4 ± 0.7 (4.8–6.0)	5.6 ± 1.1 (4.7–6.5)	0.22
MP (W)	508 ± 123 (443–574)	518 ± 124 (452–584)	0.08	480 ± 144 (360–601)	504 ± 141‡ (386–623)	0.17
MP (W·kg ⁻¹)	8.7 ± 1.0 (8.2–9.3)	8.8 ± 1.1 (8.2–9.4)	0.1	8.4 ± 1.1 (7.4–9.3)	8.6 ± 1.0‡ (7.8–9.5)	0.19
FI (%)	51.4 ± 7.0 (47.6–55.1)	48.7 ± 8.7 (44.1–53.3)	0.34	48.3 ± 6.5 (42.8–53.7)	45.5 ± 7.2‡ (39.5–51.5)	0.41
dLa/dt _{max} (mmol·L ⁻¹ ·s ⁻¹)	0.36 ± 0.06 (0.33–0.39)	0.38 ± 0.05 (0.35–0.4)	0.36	0.37 ± 0.07 (0.31–0.42)	0.37 ± 0.05 (0.33–0.42)	0
Passive						
	Pre	Post	<i>d</i>	Δ		
Ramp test						
PPO (W)	304 ± 42 (269–339)	318 ± 42‡ (283–353)	0.33			
PPO (W·kg ⁻¹)	5.2 ± 0.5 (4.77–5.61)	5.4 ± 0.4‡ (5.05–5.76)	0.44			
ṀO ₂ max (ml·kg ⁻¹ ·min ⁻¹)	56.5 ± 8.3 (49.6–63.4)	57.9 ± 8.6 (50.7–65.1)	0.17			
VT 1 _{power} (W)	176 ± 36 (146–206)	191 ± 33 (162–219)	0.43			
VT 2 _{power} (W)	231 ± 38 (199–263)	251 ± 39‡ (218–283)	0.52	§		
VT 1_ṀO ₂ (ml·min ⁻¹)	2.24 ± 0.41 (1.90–2.59)	2.39 ± 0.56 (1.92–2.86)	0.31			
VT 2_ṀO ₂ (ml·min ⁻¹)	2.67 ± 0.54 (2.22–3.12)	2.99 ± 0.46‡ (2.60–3.38)	0.64			
HR _{max} (b·min ⁻¹)	198 ± 7 (192–204)	196 ± 6 (191–201)	0.31			
Time trial						
TT (W)	169 ± 39 (137–202)	196 ± 45‡ (159–234)	0.64	(§)		
TT (W·kg ⁻¹)	2.9 ± 0.5 (2.44–3.32)	3.3 ± 0.7‡ (2.79–3.89)	0.66	§		
HR _{TT} (b·min ⁻¹)	183 ± 5 (179–188)	181 ± 7 (174–188)	0.33			
Lactate (mmol·L ⁻¹)	6.6 ± 0.8 (5.1–8.1)	8.1 ± 0.6‡ (5.9–10.4)	2.12			

Wingate			
PPO (W)	777 ± 151 (651–903)	761 ± 135 (648–874)	0.11
PPO (W·kg ⁻¹)	13.2 ± 1.2 (12.2–14.2)	12.9 ± 1.5 (11.6–14.2)	0.62
Time to PPO (s)	5.3 ± 1.4 (4.1–6.5)	5.0 ± 1.0 (4.2–5.8)	0.24
MP (W)	536 ± 99 (453–619)	532 ± 111 (439–619)	0.04 §
MP (W·kg ⁻¹)	9.1 ± 0.9 (8.4–9.8)	9.0 ± 1.3 (8.0–10.1)	0.1
FI (%)	54.5 ± 6.3 (49.1–59.8)	51.9 ± 9.3 (44.2–59.7)	0.16
dLa/dt _{max} (mmol·L ⁻¹ ·s ⁻¹)	0.36 ± 0.04 (0.32–0.40)	0.38 ± 0.04 (0.35–0.41)	0.5

*Over all = analysis of the whole group independent of the arrangement of recovery ($n = 16$); active and passive group = $n = 8$ each; PPO = peak power output; VT = ventilatory threshold; HR = heart rate; TT = time trial; MP = mean power; FI = fatigue index; dLa/dt_{max} = rate of lactate accumulation; d = Cohen's d effect size.

†Parameters are shown as mean ± SD. In brackets, 95% confidence interval is given.

‡Significantly different from prevalues. (§) Values nearly reached statistical differences (for p values see Results section).

§Significant difference between the absolute changes (Δ post-pre) of the active vs. passive group. (§) Values nearly reached statistical differences (for p values see Results section).

TABLE 3. Changes of hematocrit, Hb, and (relative) total Hb mass ([rel.] tHb mass) before (pre) and 7 days after (post) the intervention.*†

	Over all			Active			Passive		
	Pre	Post	d	Pre	Post	d	Pre	Post	d
Hematocrit (%)	41.0 ± 3.1 (39.3–42.6)	39.7 ± 2.6‡ (38.3–41.1)	0.45	40.9 ± 3.2 (38.2–43.6)	39.1 ± 2.3‡ (37.1–41.0)	0.65	41.1 ± 3.3 (38.3–43.8)	40.3 ± 2.9 (37.9–42.7)	0.26
Hemoglobin (g·dl ⁻¹)	14.1 ± 1.2 (13.5–14.8)	13.6 ± 1.0‡ (13.1–14.1)	0.45	14.0 ± 1.2 (13.0–15.0)	13.4 ± 0.8‡ (12.7–14.1)	0.59	14.2 ± 1.3 (13.2–15.3)	13.8 ± 1.2 (12.8–14.8)	0.32
tHb mass (g)	619 ± 152 (538–699)	599 ± 145‡ (521–676)	0.13	601 ± 170 (459–744)	583 ± 163 (447–720)	0.11	636 ± 140 (519–753)	614 ± 134‡ (502–726)	0.16
Rel. tHb mass (g·kg ⁻¹)	10.5 ± 1.2 (9.8–11.1)	10.2 ± 1.1‡ (9.6–10.8)	0.26	10.4 ± 1.2 (9.4–11.5)	10.1 ± 1.1 (9.2–11.0)	0.26	10.6 ± 1.2 (9.5–11.6)	10.2 ± 1.2‡ (9.2–11.2)	0.33

* d = Cohen's d effect size; Hb = haemoglobin.

†Parameters are shown as mean ± SD; in brackets 95% confidence interval is given. Over all: analysis of the whole group independent of the arrangement of recovery ($n = 16$); active and passive group: $n = 8$ each.

‡Significantly different from prevalues.

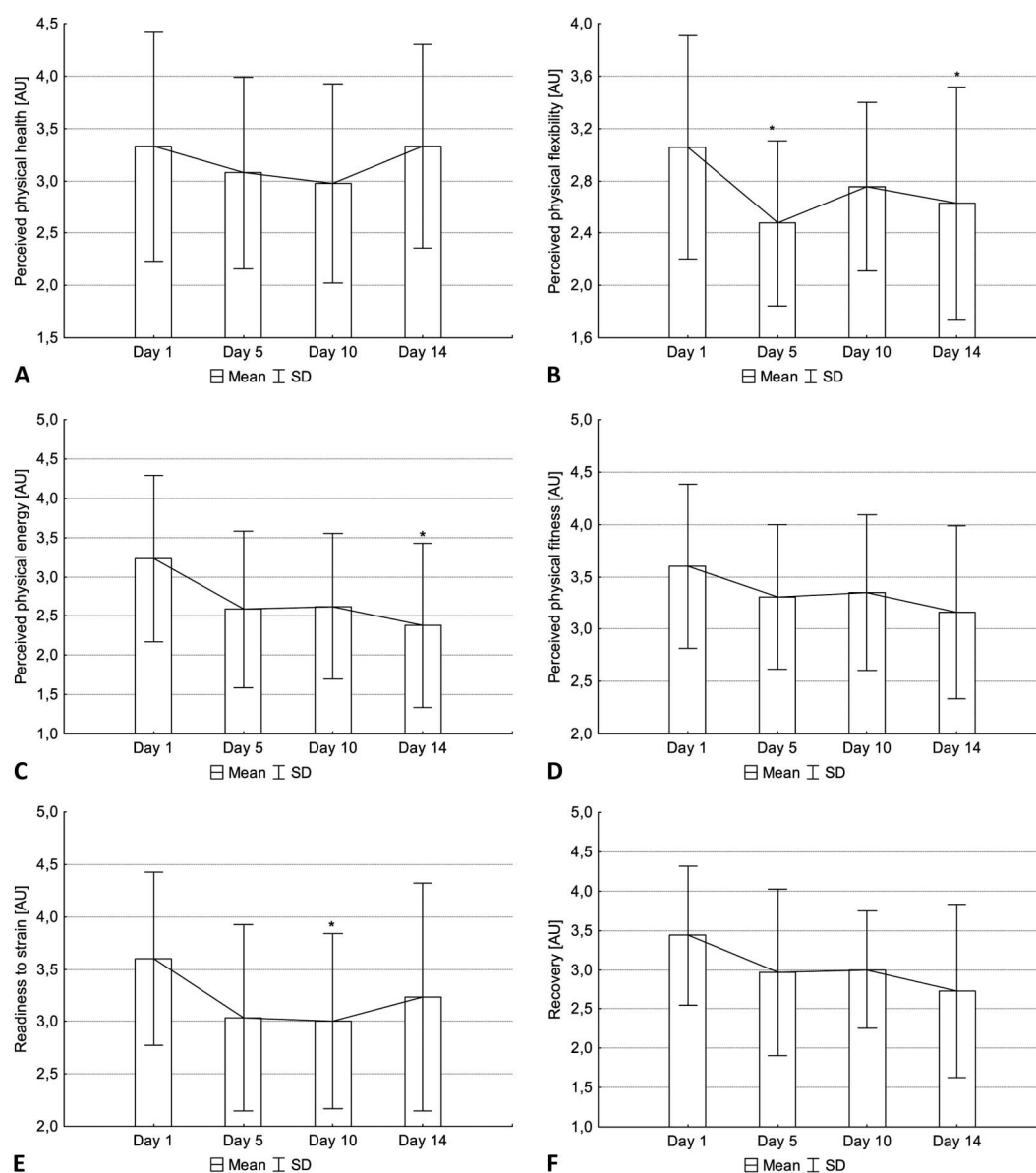


Figure 2. Changes of the person's perceived physical state (PEPS) and the psychological strain/motivational state (EZ scale). The PEPS represent 4 dimensions: perceived physical energy, perceived physical fitness, perceived physical flexibility, and perceived physical health (A–D). Of the EZ scale, only 2 dimensions are shown (E,F). Both scales were used on days 1, 5, 10, 14. Values are presented as mean $\pm$ SD. *Significantly different from day 1.

at VT2 were increased in the P group only. The comparison of absolute changes (Δ post-pre) between the A vs. P group in power output at VT2 (-8 ± 34 W [A]; $+19 \pm 15$ W [P]) revealed a significant difference between the groups (Table 2). Cronbach's α for VT1_power and VT2_power was 0.79 and 0.9, respectively; ICC: 0.65 and 0.82, respectively; Cronbach's α for VT1_ $\dot{V}O_2$ and VT2_ $\dot{V}O_2$ was 0.87 and 0.85, respectively; ICC: 0.78 and 0.74, respectively.

The PPO and relative PPO during the ramp test significantly increased in the whole group (Cronbach's $\alpha = 0.97$;

ICC: 0.94). The separate analysis of the A and P groups, showed significant increases for the P group only. However PPO, but not rel. PPO, nearly reached statistical significance in the A group ($p = 0.07$) as well. There were no statistical differences between the groups (Table 2).

Similar results were found for the MPO during the TT. The analysis of absolute and relative MPO during the TT for the whole group showed significant increases (Cronbach's $\alpha = 0.97$; ICC: 0.94). The separate analysis of the A and P groups showed significant increased for the P group only. Again,

absolute MPO, but not relative MPO, nearly reached statistical significance in the A group as well ($p = 0.053$). The increments (Δ post-pre) in relative MPO during TT ($+0.2 \pm 0.3$ W [A]; $+0.5 \pm 0.2$ W [P]) were significantly greater in the P group compared with the A group and nearly reached statistical significance for absolute MPO ($p = 0.09$) ($+14 \pm 18$ W [A]; $+27 \pm 10$ W [P]) as well (Table 2). Mean HR during TT (pre and post) reached 92% of HRmax. Mean lactate values during TT were significantly greater in the whole group and in the P group after the 2-week training period. No changes in lactate values were found for the A group.

In contrast, the A group showed significant improvements in the absolute and rel. MPO and FI during the Wingate test. No changes were found for the whole and the P group. The changes (Δ post-pre) in MPO ($+24 \pm 13$ W [A]; -4 ± 33 W [P]) during the Wingate test were significantly greater in the A group compared with the P group.

Hematocrit (Hct), hemoglobin (Hb), total hemoglobin mass (tHb mass) (Cronbach's $\alpha = 0.95$; ICC: 0.91), and relative tHb mass (rel. tHb mass) (Cronbach's $\alpha = 0.99$; ICC: 0.98) significantly decreased in the whole group. The separate analysis of the 2 groups showed significant decreases for Hct and Hb in the active group and significant decreases for tHb mass and rel. tHb mass in the P group (Table 3). No significant differences were found for absolute changes (Δ post-pre) between the groups.

The perceived physical energy, perceived physical flexibility, readiness to strain and sleepiness significantly decreased during the 2 weeks (Figure 2). The perceived physical fitness, perceived physical health, and recovery did not change (Figure 2).

DISCUSSION

The main findings of this study are (a) that a 14-day shock microcycle including 15 HIT sessions is able to improve time-trial performance and PPO in junior triathletes in a short period of time without concurrent changes in $\dot{V}O_{2\max}$ and (b) that passive recovery is superior compared with active recovery in inducing improvements in absolute and relative TT performance, and power output at VT2, despite a slightly lower training volume. By design, the absolute training volume was 20% lower in the passive group (8.9 vs. 11 hours) compared with the active group, because of the resting phases between the intervals. Therefore, this study gives a first hint of the importance of the arrangements of interval training, especially the planning of load-rest intervals; however, the potential (cellular) mechanisms for the different effects of active and passive recovery need to be identified in future studies.

Previous studies showed that HIT can induce the improvement of $\dot{V}O_{2\max}$ and endurance performance, but little was known about its usage during various training periods in terms of single HIT sessions per week or intensifications of whole weeks (shock microcycle). In accordance with the results of Breil et al., this study underlines the effectiveness of

a HIT shock microcycle to improve endurance performance (2). Although we were not able to show major improvements of $\dot{V}O_{2\max}$ because of the shock microcycle like in the studies of Breil et al. (+6%) (2) and Stolen et al. (+7.3%) (34), TT performance significantly increased by +12% in this study. Previous studies with well-trained cyclists already showed that HIT can improve TT performance without the detection of concurrent increases in $\dot{V}O_{2\max}$ (19,35). Because not all improvements of performance might be detectable in improvements of physiological measures and parameters, the measurement of performance should be the main criterion for the efficiency of a training intervention. However these studies “only” performed approximately 2 HIT sessions per week during a 2- to 6-week training period and used different HIT protocols (19,20,23,35) (for review see [37]). We showed that these improvements can also be induced with a higher density of HIT sessions in shorter period of time. In contrast, Breil et al. were not able to show improvements in performance in terms of time to exhaustion (t_{lim}) and total number of jumps performed in 90 seconds in a ski-specific high-box jump test (2). However, this might be because of the chosen testing procedures, as time-to-exhaustion (t_{lim}) protocols show a higher coefficient of variation (CV) of >25% in contrast to TTs (<5%) (7), and it is more difficult to measure the impact of an athletic training intervention on ski-specific performance than in endurance disciplines.

As total hemoglobin mass (tHb mass) is a key determinant for aerobic capacity with strong positive correlations between tHb mass and $\dot{V}O_{2\max}$ in adults (31), we expected both parameters to increase. The tHb mass related to body weight was similar to the values reported by Ulrich et al. (36) in endurance trained (11.2 ± 1.6 g·kg⁻¹) and non endurance trained athletes (9.7 ± 1.3 g·kg⁻¹) of a similar age (15–17 years) (36). Ulrich et al. showed that tHb mass related to body weight did not change significantly after 6, 12, and 18 months of regular training. In contrast, tHb mass and relative tHb mass significantly decreased in this study. This small but significant decrease might be because of the high impact of HIT causing a loss of red blood cells, which cannot be compensated in this short period of time. However, the decrease of tHb mass and relative tHb mass did not affect TT performance but might explain the absence of increases of $\dot{V}O_{2\max}$.

Besides the limited knowledge of the effectiveness of HIT shock microcycles, only a little was known about the arrangement of interval training (active vs. passive recovery) and especially about the long-term effects of different arrangements. In most cases, rest periods during interval training are carried out actively, performing low-intensity exercise to reestablish performance and the homeostasis of the cell (to decrease lactate levels and to normalize pH) faster compared with passive rest (12). Furthermore, Hoff et al. (16) favored active recovery between bouts as it “prevents” a marked drop in HR and stroke volume (SV)/cardiac output (Q), allowing to reach the intended values of HR (90–95% of HRmax) and SV and therefore the intended stimulus faster in the

following bouts (16). However, this study showed advantages of passive recovery in improving endurance performance, in terms of TT performance and $\dot{V}O_2$ /power output at VT2. As metabolic alterations are essential for muscle adaptations in endurance training (4), passive recovery might lead to greater and more long-lasting metabolic alterations and therefore to greater adaptations and improvements of performance. Passive recovery leads to more severe muscle fatigue and might increase the time of exposure to potent metabolic stimuli like lactate and pH to the (muscle) tissue, possibly leading to different (adaptation) responses compared to active recovery. For example, previous studies showed that changes in the acid-base balance affect the hormonal response to exercise (30,39) and that microenvironment signals, such as acidosis and lactate (which are more severe after passive recovery) play a major role in the control of vascular endothelial growth factor production, and consequently in modulation of angiogenesis (9,10). Even though we did not measure cellular adaptations, the greater disturbances of homeostasis because of passive recovery might explain its advantage.

However, the arrangement of HIT, especially active vs. passive recovery, might also influence the general demand. The high demand and exhausting character of the intervention was shown by significant decreases in some dimensions of the PEPS scale. A large change or increase in training load, like its present during a HIT shock microcycle may lead to overreaching or even to overtraining (3). A short-term overreaching might be acceptable, possibly leading to an amplified supercompensation and increases in performance. However, as Breil et al. (2) stated, it cannot be determined to what extent we reached or possibly overreached the limit of tolerable training load using a 14-day HIT shock microcycle (2). Therefore, the screening of perceived physical state might be a useful and supporting tool in such intense training interventions, to get further information about the athletes (perceived) physical state. But the question remains, how many HIT sessions are tolerable.

PRACTICAL APPLICATIONS

This study showed that a 14-day shock microcycle can be included in the preparation period and is able to improve TT performance and PPO in junior triathletes in a short period of time. Furthermore, the present results give a first hint at the importance of the arrangements of interval training, especially the planning of load-rest intervals, as passive recovery induced greater improvements of endurance performance than active recovery; however, the results need to be confirmed by future studies and the potential (cellular) mechanisms for different effects need to be identified. According to the present data, one can speculate that reactions and adaptations might vary according to the arrangement of recovery. Active recovery might be useful to improve SV/Q and for recovery reasons, passive recovery might elicit greater disturbances of homeostasis and therefore a higher (local) stimulus in skeletal

muscles leading to increased local adaptations and greater improvements. Therefore, the coach should carefully choose active or passive recovery, depending on the aim of the training intervention.

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